University of Cape Town

Supervised Learning Assignment 1

Boston Median House Values Prediction

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Introduction

This report explores and analyzes the key predictors of median house values in the Boston area using various linear regression techniques. The predictors range across multiple factors, including the average number of rooms per household (rm), the concentration of nitrous oxide pollutants in the area (nox), and the full value of property taxes (taxes). The regression results of these various models are compared against each other to gauge which models and predictors capture the variation of median house prices observed in the data. The key contention among the linear models is the balancing of out-of-sample accuracy as well as the inferential and interpretable soundness of the models.The presence of multicollinearity among predictor variables presents a key challenge to understanding the marginal contribution of the individual regressors to the response variable, medv.

To address this, several variable selection approaches are compared to the baseline: **Backward Stepwise Regression, Ridge regression, LASSO Regression**, and the inclusion of an **Interaction-Feature-Included** model. Model performance is evaluated using both mean squared error metrics, residual diagnostic checks, as well as the significance and magnitudes of the various predictors across the different models (and the stability thereof). Finally, important consideration is given to the trade-off between model complexity and interpretability, and the justification for which model performed best, given all the aforementioned information.

Exploratory Data Analysis

For the sake of data and analysis completeness, EDA is conducted on the entire Boston dataset.

Data Attributes

Table 2: Boston Data Data Summary

feature	min	q25	mean	meadian	q75	max	sd	datatype	na_count
crim	0.00632	0.082045	3.6135236	0.25651	3.677083	88.9762	8.601744	numeric	0
zn	0.00000	0.000000	11.3636364	0.00000	12.500000	100.0000	23.322521	numeric	0
indus	0.46000	5.190000	11.1367787	9.69000	18.100000	27.7400	6.860029	numeric	0
chas	0.00000	0.000000	0.0691700	0.00000	0.000000	1.0000	0.244949	integer	0
nox	0.38500	0.449000	0.5546951	0.53800	0.624000	0.8710	0.100000	numeric	0
rm	3.56100	5.885500	6.2846344	6.20850	6.623500	8.7800	0.700000	numeric	0
age	2.90000	45.025000	68.5749012	77.50000	94.075000	100.0000	28.148890	numeric	0
dis	1.12960	2.100175	3.7950427	3.20745	5.188425	12.1265	2.104757	numeric	0
rad	1.00000	4.000000	9.5494071	5.00000	24.000000	24.0000	8.707468	integer	0
tax	187.00000	279.000000	408.2371542	330.00000	666.000000	711.0000	168.537118	integer	0
ptratio	12.60000	17.400000	18.4555336	19.05000	20.200000	22.0000	2.165641	numeric	0
lstat	1.73000	6.950000	12.6530632	11.36000	16.955000	37.9700	7.140728	numeric	0
medv	5.00000	17.025000	22.5328063	21.20000	25.000000	50.0000	9.197282	numeric	0

The complete data set contains 506 observations across 13 columns. The data is complete, with no null values, and all columns are registered as numeric. **Tax (28404), age (792), and zn (543)** display the greatest variance across the features, whilst **nox (0.01) and rm (0.49)** show the least variance across their respective values.

Median House Value Distributions

The shape of the distribution of median house values is fairly bell-shaped, with 41% (208) of median house values falling between approximately $18K and $23K. While 93% (471) of observations belong to the category with `chas = 0`, the median house value for observations with `chas = 1` is 11% higher at $23.3K compared to those in the `chas = 0` category. Additionally, observations in the `chas = 1` group show a skew towards higher values, unlike those in the `chas = 0` group, which exhibit a more balanced distribution around the median. Despite these differences in distribution, the two boxplots overlap, suggesting that the observed difference in means between the two groups may not be immediately statistically significant[1].

[1] A difference-of-means hypothesis test would need to be conducted to formally assess whether the observed difference between group means is statistically significant.

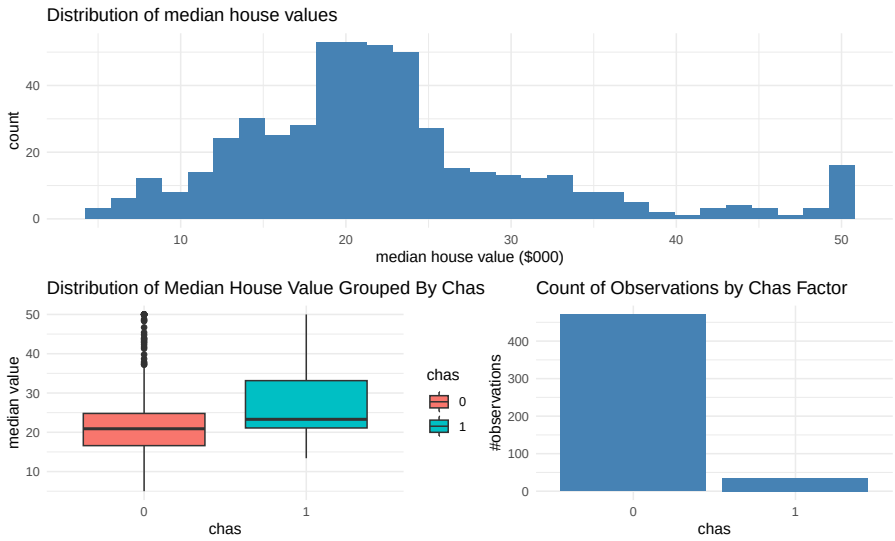


Figure 1: Median House Value Distribution,Grouped by Chas

Figure 2: Distribution of Median House Values by Radial Accesibility Index

Most observations fall within the RAD index values of 4, 5, and 24. When group-

ing house value distributions by RAD index value, there is no discernible trend in median house values with changes in the RAD index. However, observations with a RAD value of 24 show the lowest house values among the different index, with no overlap with the other box plots.

Feature Correlations

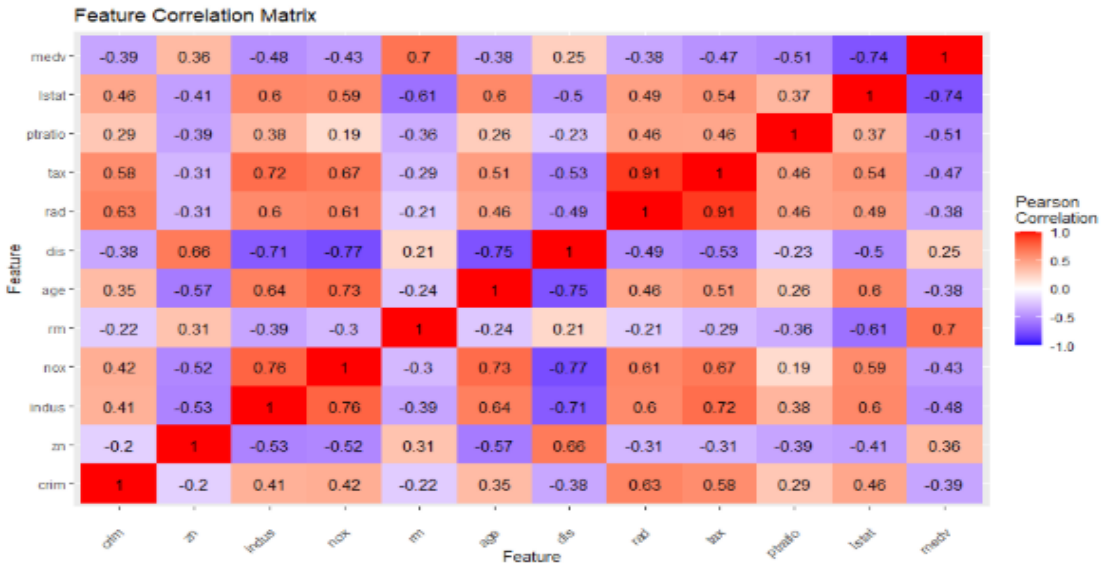


Figure 3: Feature Correlation Heatmap

The data shows evidence of multicollinearity amongst regressors, with 8 out of 11 explanatory features reporting linear correlations >=70%. In particular, features, **rm and tax** display the largest correlation (~91%), in absolute value. Features, **lstat (~-74%)** and **rm (~70%),** show the highest absolute value correlations to the response value variable.

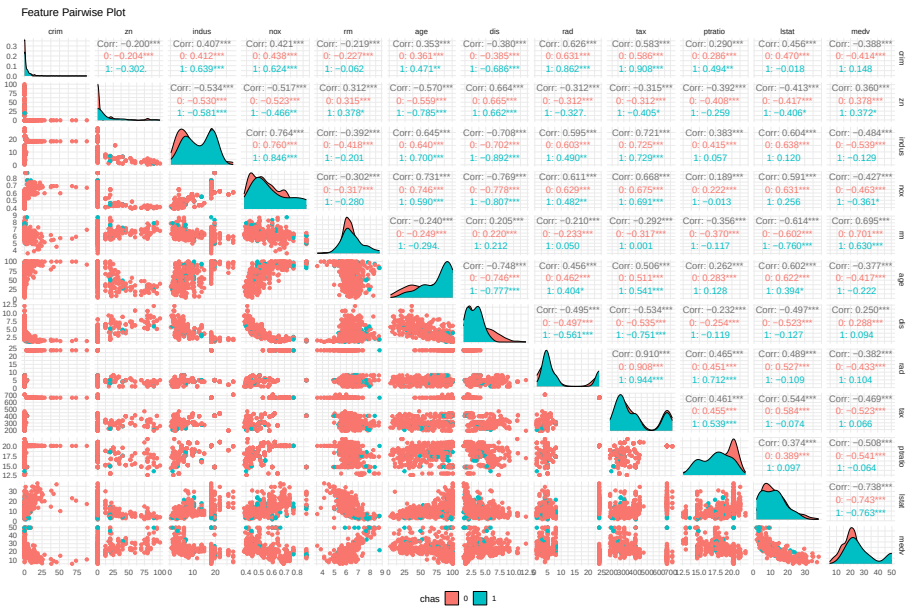


Figure 4: Feature Pairplot

The correlations between the response variable, **medv**, and all other numeric features are significant at the 1% level. The observed relationship between medv and its highest correlated feature, **lstat** appears to be negatively non-linear upon visual inspection of the shape of the scatter plot.

Grouped by the **chas** feature, for cases where chas =1 (which have shown to have higher median house values) median house values correlations lose their statistical significance across 7 features, namely **crime, indus, age ,dis, rad, tax, and ptratio.**

Another interesting feature interaction is that of the **nox** and the **dis** features, which show a significant negative correlation, with the shape of the scatter plot also indicating non-linearity. This is intuitively reasonable as the greater the distance to employment centers, the less traffic in the area, resulting in lower levels of pollutants (nox).

Model Building

Baseline Model

Method: Use all regressors in the model, with minimal transformation. The **chas** variable will be transformed into a factor as it is categorical whilst the rad (index) variable will be kept as a numeric variable to preserve its ordinality

Characteristic	Beta[1]	SE
(Intercept)	44***	5.65
crim	-0.13**	0.039
zn	0.06***	0.016
indus	0.03	0.077
chas		
0	—	—
1	1.8	0.957
nox	-21***	4.41
rm	3.7***	0.486
age	-0.01	0.015
dis	-1.7***	0.233
rad	0.29***	0.080
tax	-0.01**	0.005
ptratio	-1.0***	0.147
lstat	-0.48***	0.058

[1]*p<0.05; **p<0.01; ***p<0.001
Abbreviations: CI = Confidence Interval, SE = Standard Error
R^2 = 0.738; Adjusted R^2 = 0.730; No. Obs. = 400
Fstat (12, 387) = 90.99***, p = 1.19e-104

Figure 5: Baseline Model Regression Summary

Model Performance:

- The model shows a relatively good fit to the training data, with ~74% of the variation in the data, explained by the model
- 9 out of the 12 feature coefficients (excluding the intercept) are statistically different from 0 at at lest the 10% level, with 9 of these significant at the 1% level
- Nitrogen Oxide concentration (nox) levels show the biggest marginal contributer, in absolute value, to median house values, medv, with an estimated negative change of -$20.6K in estimated average house values per one unit increase in nox levels.
- The F statistic is also significant at the 1% level, indicating that the coefficients of the model are jointly statistically different from 0.

Prediction of test data set

Table 1: Baseline Test MSE

baseline.mse
16.64

The out-of-sample (OOO) MSE for the baseline model is $16.64K, that is, on average the baseline model's home values predictions are expected to deviate from their actual by this values (which is greater that the standard deviation of median house values of ~$9.2K).

Model Improvement

Backwards Step wise Variable Selection

Method:

1. Start off with the baseline model: regress response variable against all features
2. Iteratively remove one feature, retrain the model and store both the AIC and BIC scores for each retrained model- the StepAIC function terminates when subsequent feature removals does not improve AIC/BIC.

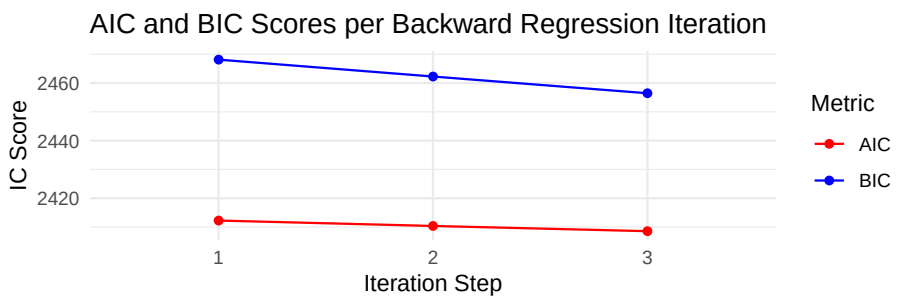


Figure 6: AIC and BIC curves per Backwise Steo Regression Iteration

Characteristic	Beta[1]	SE
(Intercept)	44***	5.61
crim	-0.13**	0.039
zn	0.06***	0.016
chas		
0	—	—
1	1.8	0.948
nox	-20***	4.05
rm	3.7***	0.477
dis	-1.6***	0.218
rad	0.28***	0.075
tax	-0.01***	0.004
ptratio	-1.0***	0.146
lstat	-0.49***	0.054

[1]*p<0.05; **p<0.01; ***p<0.001
Abbreviations: CI = Confidence Interval, SE = Standard Error
R^2 = 0.738; Adjusted R^2 = 0.731; No. Obs. = 400
Fstat (10, 389) = 109.63***, p = 1.26e-106

Figure 7: Backward Stepwise Regression Summary

Backward Step Regression Results:

1. The step model reached the lowest AIC (~2408) and BIC(~2456) after 3 iterations with 10 out of the 12 features retained (subsequent removals of features did no improve scores).
2. Features **crime, zn, chas, nox, rm, dis, rad, tax, ptratio and lstat** are retained whilst the **age** and **indus** variables are removed**.** All of these are still statistically significant at least up to the 10% level.
3. All regressors also maintain their sign with very little changes to their marginal effects with respect to the response variable (realtive to the baseline model).
4. R square (~74%) and the F statistic (significant at 1%) are still in the same range and region as the baseline model.

Ridge and Lasso Regression

Method: Use cross validation to find the optimal regularization parameter(s) ,λ, that produce the minimum CV MSE error during training and compare the coefficients produced by the model.

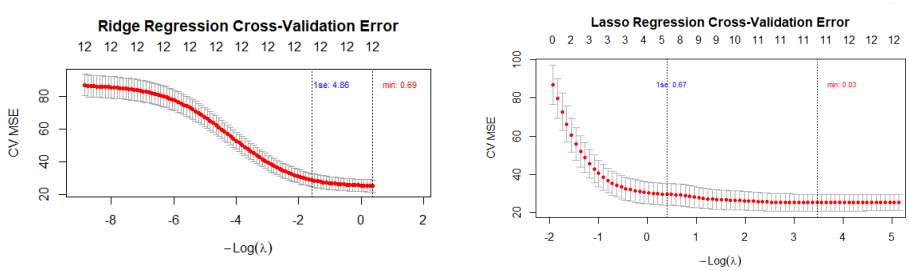


Figure 8: Cross-validation mean squared error plots for Ridge and Lasso regression.

Table 2: Ridge and LASSO Regression Coefficients

Feature	Ridge λ_{min}	Ridge λ_{1se}	LASSO λ_{min}	LASSO λ_{1se}
(Intercept)	34.7669663	25.4992007	42.5668194	20.4436614
crim	-0.0997802	-0.0704068	-0.1185652	-0.0232736
zn	0.0403742	0.0248068	0.0520495	0.0000000
indus	-0.0420514	-0.0746080	0.0000000	0.0000000
chas	2.0742555	2.1120254	1.7878073	0.7975187
nox	-13.8315658	-5.8133328	-19.3000339	-3.8251303
rm	3.9794124	3.4638733	3.7861161	4.3258353
age	-0.0082776	-0.0113347	-0.0039237	0.0000000

Feature	Ridge λ_{min}	Ridge λ_{1se}	LASSO λ_{min}	LASSO λ_{1se}
dis	-1.2318078	-0.4766797	-1.5880810	-0.1559487
rad	0.1208266	-0.0006445	0.2381425	0.0000000
tax	-0.0071776	-0.0037944	-0.0122039	-0.0020665
ptratio	-0.9140152	-0.6941726	-0.9950010	-0.8503920
lstat	-0.4340762	-0.3124781	-0.4792334	-0.4635408

Regression Results:

1. The best regularization parameters for the Ridge and LASSO models are reached at at λ =0.69 (CV_MSE$_{\text{min}}$ of ~25.48) and 0.03 (CV_MSE$_{\text{min}}$ ~27.9), respectively . The 1_{se} equivalent parameters for Ridge and LASSO are obtained at λ=4.86 (CV_MSE$_{\text{min}}$ ~28.79) and λ=0.67 (CV_MSE$_{\text{min}}$ ~27.99).
2. For the simplest model, LASSO$_{\text{1SE}}$, 7 non-zero coefficients remained in the final model; the coefficients of **zn, indus, age,** and **rad** were all pushed to zero. In particular, **zn and rad** were statistically significant in the baseline model.
3. The **nox** feature still maintains its position as the largest absolute marginal predictor of median house values across three out of the four models, with the **rm** feature having the largest marginal contribution to response variable in the LASSO 1_{se} model.
4. In addition to a change in magnitude, some feature coefficients also show a change in sign in some of the models: the **indus** coefficient changes from positive to negative from the baseline in both the Ridge$_{\text{1SE}}$ and Ridge$_{\text{min}}$ models while the **rad** feature coefficient changes sign in the Ridge$_{\text{1SE}}$ model (although this feature coefficient is highly penalized relative to the baseline).

Predictions On Test Data

For both the Ridge and LASSO regressions, the 1SE models will be used to favour model simplicity and further reduce the risk of over fitting; these are used to predict on the test data.

Table 3: Test Data MSE Model Metrics

metric	all_features	back-wise_step	ridge	lasso	average.across.models
MSE	16.64	16.65	16.96	19.77	17.51

Although the test scores MSE are relatively close to each other for 3 out of the 4 models, the variable selection models fail to outperform the baseline model, with the Backward Stepwise model coming in close second. The LASSO model

has the worst test MSE out of the 4 at \$19.77K-11% higher than the average across models.

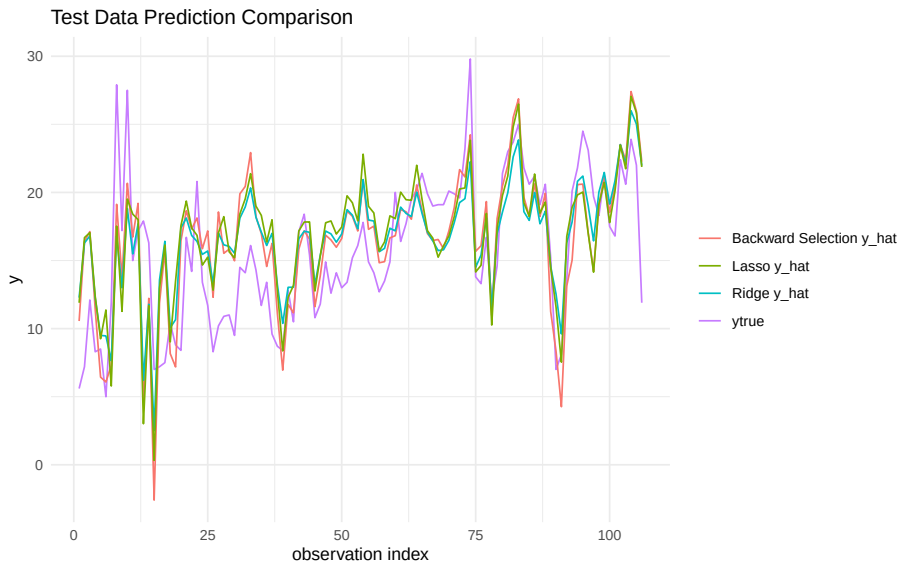


Figure 9: Predictions Vs Actual Values

All the model predictions show relatively good fit to the general shape and trend of the response variable in the test data, although there is general under prediction at some extreme points of the response data.

Inclusion of Interaction Term

Method: The **chas** and **rad**[2] categorical variable provides an easy way of segmenting the data into distinct groups for which interactions with other variables can be created and analyzed. Thus the following process is used to choose the interaction feature:

1. Create interaction terms for all features with the chas feature: $ChasIntVar = \text{Feature}_i \times \text{chas}$
2. Do the same for the rad feature: $RadIntVar = \text{Feature}_i \times \text{rad}$
3. Analyze the correlations of all these interaction variables with respect to the response variable, and choose the interaction with the highest, and most statistically significant correlation.
4. Add this term to the best-performing model, which in this case was the **full variable regression.**

[2]Measured as an index as per meta data definition

5. Again, for the sake of data completeness, the interaction correlation analysis is done on the full data set before the addition to the model.

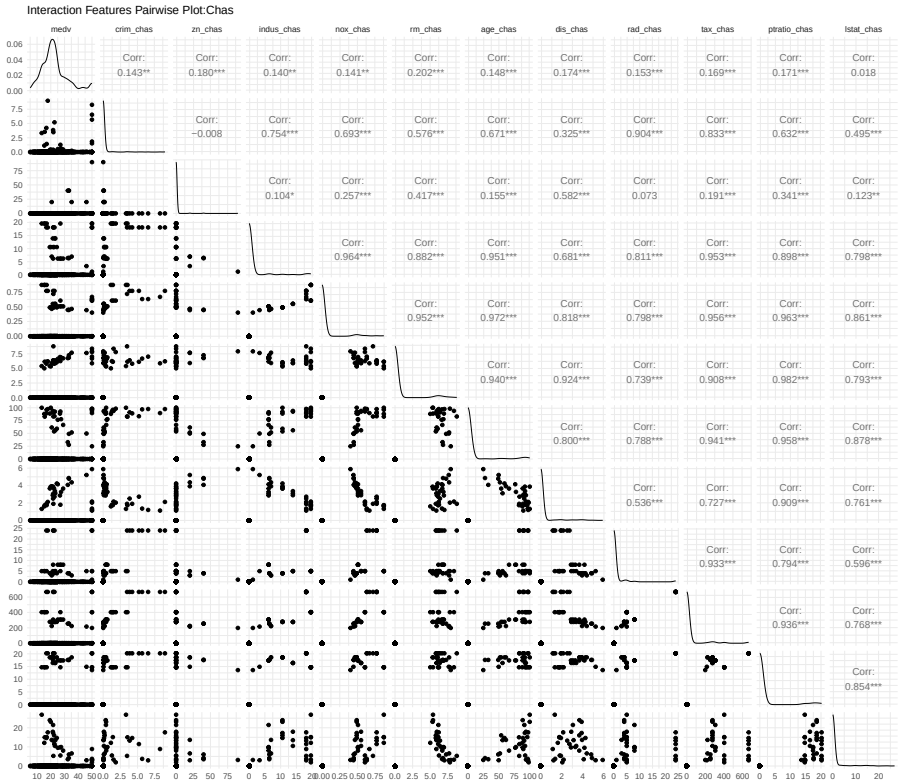


Figure 10: Interaction Features Pairplot-chas feature

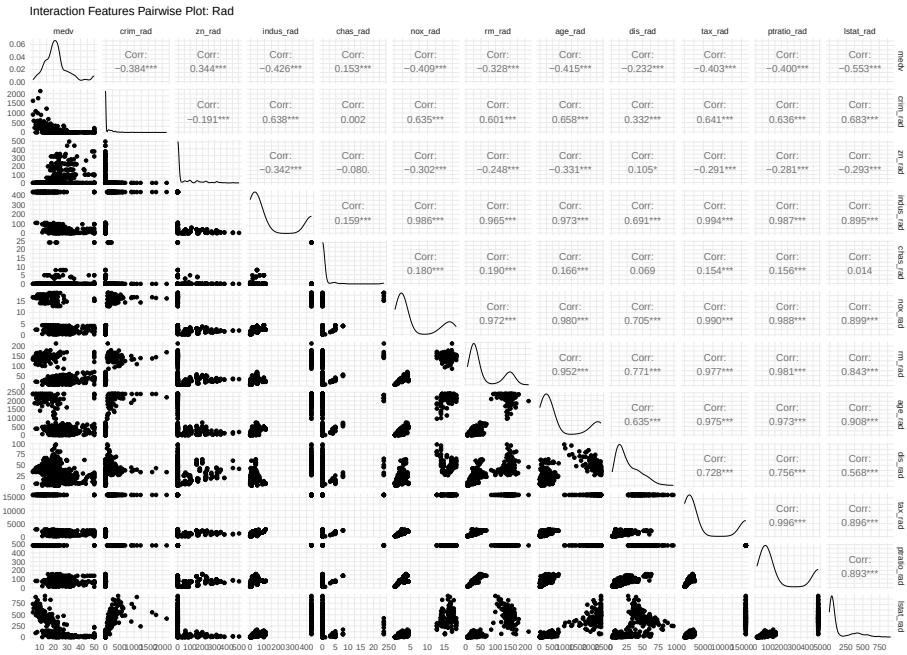


Figure 11: Interaction Features Pairplot- rad feature

Across both pair plots, the **lstat_rad** interaction feature show the highest (in absolute value) and most statistically significant correlation (at 1%) among interacted features. This will be added to the training data for the best performing model.

Characteristic	Beta[1]	SE
(Intercept)	44***	5.67
crim	-0.12**	0.040
zn	0.06***	0.016
indus	0.02	0.078
chas	1.7	0.960
nox	-21***	4.40
rm	3.8***	0.490
age	-0.01	0.015
dis	-1.7***	0.236
rad	0.38***	0.111
tax	-0.01**	0.005
ptratio	-1.0***	0.148
lstat	-0.40***	0.085
lstat_rad	-0.01	0.005

[1]*p<0.05; **p<0.01; ***p<0.001
Abbreviations: CI = Confidence Interval, SE = Standard Error
R^2 = 0.739; Adjusted R^2 = 0.731; No. Obs. = 400
Fstat (13, 386) = 84.20***, p = 5.6e-104

Figure 12: Interaction-Variable-Included Regression Summary

Despite the significance of the two individual features, **lstat and rad**, the

engineered interaction feature did not prove to be a statistically significant predictor of median house values.

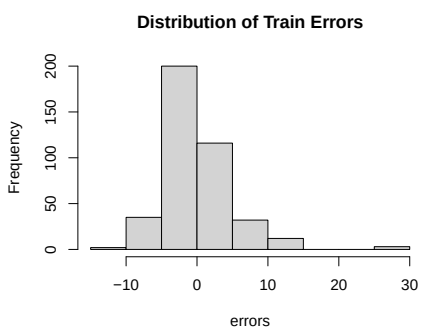
Table 4: Test Data MSE Comaparison: Baseline vs Interaction -Included Model

Metric	Baseline	Interaction
MSE	16.6422	16.78386

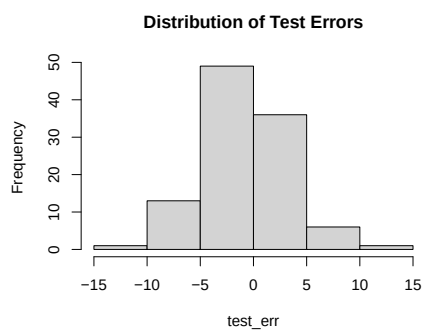
Again, the inclusion of the interaction did not improve model performance, although the model mse deterioration was marginal compared to the test MSEs of LASSO and Ridge regressions.

Residual Diagnostics

Residual analysis is conducted on the "best" performing model; the all-feature model.



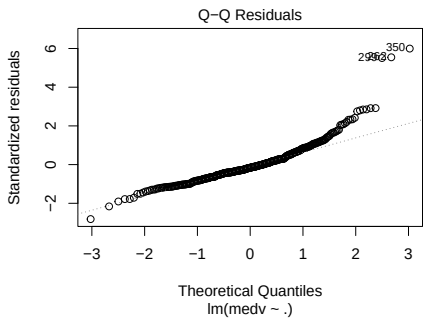
(a) Train Residuals



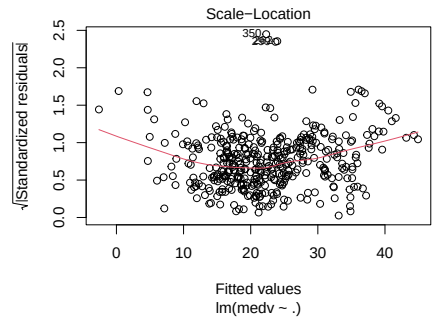
(a) Test Residuals

Despite being the best performing model, residual analysis of the baseline model does not indicate that the errors follow the assumptions of the linear model, namely:

1. Normality: Although the histograms for both the train and test residuals may indicate normality, the standardized residual points on the QQ plot do not fall on the straight line when compared to the theoretical normal distribution quantiles, suggesting non-normality. This greatly affects the ability and validity of statistical inference, especially concerning hypothesis testing on the key regression statistics, including prediction confidence intervals of the response variable (although the estimated coefficients are mostly unaffected).



(a) QQ Plot To Check for Normality



(a) Residual Variance Plot to check Homoscedascity

2. Homoscedasticity: The scale-location plot does not indicate that the variance of the residuals is constant[3], rather it seems to follow an increasing trend when plotted against fitted values. This also greatly inhibit the ability to make valid statistical inferences, especially regarding the variance and precision of the regressors as well as the overall validity (significance) of the model.

Conclusion: Considerations and Implications of the "Best" Model

Model accuracy and inference play a significant role in selecting the appropriate model for a specific task, and no single metric wholly captures the overall performance of a model. Although the baseline model reported the best out-of-sample performance based on the test Mean Squared Error (MSE), its observed model diagnostics were inadequate; challenging efficacy and consistency of the model if inference is the ultimate objective.

Multicollinearity is a key attribute of the data(which the baseline model cannot address without significant modification to the features) that significantly impacts the ability to disentangle the marginal influence of individual predictors on the response variable(K 2023). Ridge regression effectively penalizes the coefficients of highly correlated predictors (tax and rad) with a negligible impact on out-of-sample accuracy.

Additionally, increasing model complexity through the addition of the interaction term did not improve accuracy; the term itself proved to be a statistically

[3]Homoscedasticity requires the plotted points to be scattered randomly about the line, with no discernible pattern.

insignificant predictor of house values. Across the models, **pollutant levels (nox)** and the **average number of rooms per dwelling (rm)** emerged as the most significant marginal predictors of house values, consistently maintaining their rank and sign.

Given these considerations, the Ridge regression model is the most appropriate choice, as it strikes a favorable balance between maintaining model accuracy, simplicity, and interpretability.

Appendix

Table 5: Baseline Model: Feature Variance Inflation Factor

	VIF
crim	1.845656
zn	2.231267
indus	4.575643
chas	1.083827
nox	4.556684
rm	2.007421
age	3.048920
dis	3.951907
rad	8.074803
tax	10.351774
ptratio	1.694486
lstat	2.951298

All features moderate to high variance inflation factors, further supporting evidence of multicollinearity among features, with the biggest contributors are being the tax and rad features.

References

K, Boriharn. 2023. "Visualizing the Effect of Multicollinearity on Multiple Regression Model." *TDS Archive*, June. https://medium.com/data-science/visualizing-the-effect-of-multicollinearity-on-multiple-regression-model-8f323ef542a9.